

Fantine Mpacko Priso

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Profile

Master's student in Computer Science and Mathematics, (expected graduation in December 2026), with research experience in machine learning optimization, statistical modeling, computer vision, and human-centered AI. Proficient in Python, MATLAB, C++, Java; Familiar with SQL, HTML, JavaScript, CSS and R.

Education

University of California, Berkeley

2025–2027

Graduate Student

- Graduate coursework in Machine Learning, Deep Neural Networks Architectures, Optimization, Stochastic Processes, Computer Vision, Functional Analysis.
- Computational precision health & neuronal function coursework.

Paris-Saclay University, Ecole Normale Supérieure Paris-Saclay, France

2021–Present

MSc in Computer Science (current), Double Major BSc in Mathematics and EECS (graduated June 2024)

- Graduate coursework in Reinforcement Learning, Signal processing, multidimensional system, and information transmission.
- Coursework in parametric identification and optimization, probability theory, digital electronics, signal processing, linear algebra, real analysis.

Research Experience

Research Intern — Computational Cognitive Neuroscience

Summer 2026

Cognitive & Computational Neuroscience Lab (LNC2), Ecole Normale Supérieure, Paris

- Developed and deployed a JATOS-based behavioral experiment with three study conditions comparing face and grayscale-circle proportion estimation, including randomized trials, response timing, pilot debriefing, and structured data exports.
- Implemented participant-specific face reconstruction using Bradley-Terry preference modeling, MediaPipe landmarks, geometric alignment, and Delaunay-based morphing.
- Built a reproducible Python stimulus pipeline, auditing 597 Chicago Face Database records and selecting an age-matched set of 122 faces to generate 45 paired face-circle matrices.

Selected Projects

Modular Manifolds in Practice: Accelerating MuOn Optimizer with Quasi-Newton Methods

- Developed a fixed-step Riemannian L-BFGS inner solver with cautious curvature updates for the Muon optimizer.
- Reduced per-epoch runtime from approximately 110 s to 25 s ($4.4\times$ speedup) on a CIFAR-10 MLP while maintaining test accuracy within 4–6 percentage points of the original dual-ascent Muon solver.
- Benchmarked the method across MLPs, CNNs, and a small Vision Transformer on CIFAR-10 and CIFAR-100 using learning-rate sweeps and up to four random seeds.

Same Symptoms, Different Advice: An Intersectional Audit of AI Mental Health Support

- Designed an audit study to evaluate whether AI mental health support systems provide different advice across demographic and intersectional user profiles.
- Developed matched scenarios that held symptoms constant while varying demographic attributes to isolate demographic and intersectional effects.

EEG-Guided Visual Stimulation System

- Analyzed 16-channel EEG recordings sampled at 500 Hz to characterize occipital brain activity and identify alpha-band oscillations around 11 Hz.
- Implemented a third-order Butterworth filtering and signal-quantization pipeline in Python to translate EEG dynamics into discrete stimulation patterns.

Libraries & Tools:

NumPy, Pandas, Polars, SciPy, MediaPipe, Streamlit, PyTorch, scikit-learn, CUDA, OpenCV, Pillow, Git and version control, Google Apps Script, JATOS.

Leadership and Community Engagement

- Represented a community of approximately 1,000 students and managed critical stakeholder communications, organizing events with budgets up to €100K and attendance of up to 600 participants.
- Co-founded and led UFO.recordz, managing events with budgets up to €15K, audiences of up to 2,500 participants, and social-media reach of up to 500K monthly views.